



Review article

Hybridization of stochastic hydrological models and machine learning methods for improving rainfall-runoff modeling

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ABSTRACT

Accurately simulating river discharge remains a challenge. Hybrid models combining hydrological models with machine learning improve discharge simulation and offer better interpretability than standalone machine learning models. However, the commonly used models are deterministic. This study introduces an innovative extension to stochastic hydrological models, offering a novel combination that has not been previously explored. The proposed approach predicts discharge by integrating the simulated statistical properties of daily discharge probability distributions, derived from a stochastic rainfall-runoff model, into machine learning frameworks. This integration allows the machine learning models to incorporate insights from the uncertainties in discharge, thereby enhancing predictive accuracy of discharge simulations. The hybridization presented combines the physically-based stochastic HyMoLAP (Sto. HyMoLAP) model with machine learning techniques, including Wavelet-based eXtreme Gradient Boosting (WXGBoost) and Wavelet-based Gated Recurrent Unit (WGRU). Evaluated on the Ouémé at Bonou river basin, Benin, the Sto. HyMoLAP-WGRU model shows the best predictive performance, especially for low and high discharges. It achieves an overall Nash-Sutcliffe Efficiency (NSE) of 0.896, which is 7.30% higher than the NSE of HyMoLAP, and 29.67% and 259.71% higher than those of the standalone machine learning models. The Combined Accuracy (CA) is 38.11, reflecting reductions of 19.81%, 42.30%, and 62.41% compared to the standalone models. The analyses show that the performance of hybrid models depends on the simulated discharge distribution properties used as input. They suggest that the hybridization approach could be particularly beneficial for runoff simulations in catchments subject to significant random fluctuations where point discharge simulation is challenging.

1. Introduction

Simulating streamflow over extended periods with precision is crucial for water resource management, extreme events forecasting, and environmental conservation. However, many hydrological models struggle to capture the inherent variability within river basins. Recent advancements in machine learning offer new opportunities to enhance these models by integrating data-driven techniques with conceptual or physically-based hydrological approaches, resulting in more robust and precise discharge simulations. These hybrid approaches capitalize on the capability of machine learning models to better capture non-linear hydrologic responses as compared to process models [1–3]. Conceptual models, often referred to as gray-box models, are rainfall-runoff

models developed using observed or hypothesized empirical links between various hydrological variables. Physically-based models are based on the physical process, which includes continuity, momentum, and energy principles [4]. They refer to distributed or semi-distributed hydrological models, which differ from conceptual models by relying on physical equations to represent the spatial characteristics of fluid flow within a catchment [5]. Data-driven models are computational frameworks that rely on statistical techniques, such as the ARIMA (Autoregressive Integrated Moving Average) model, as used for streamflow simulation in [6], and machine learning algorithms to analyze and learn patterns from historical data. Unlike conceptual and physically-based models, data-driven approaches do not explicitly simulate hydrological processes. Instead, they are based on the relationship be-

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tween input and output data without considering the physical process [7].

Several conceptual and physically-based hydrological models have been used for rainfall-runoff modeling. Particularly, hydrologists have paid attention to developing physical-based models, such as rainfall-runoff (R-R) models, to simulate streamflow patterns [8]. In the past few decades, there have been extensive endeavors to create, validate, and compare various physically-based hydrological models designed for rainfall-runoff modeling across diverse regions [9]. The physically-based models include the Variable Infiltration Capacity (VIC) model [10], the European Hydrological System-Système Hydrologique Européen (SHE) [11], the Soil and Water Assessment Tool (SWAT) model [12], and the Hydrological Model based on the Least Action Principle (HyMoLAP), which is essentially an Ordinary Differential Equation (ODE), and whose theoretical development was initially elaborated by [13–15]. In fact, the HyMoLAP model's formulation relies on the Least Action Principle (LAP), a fundamental concept in physics that has been significantly enriched by Noether's theorem [16], which links symmetries in the action to conservation laws. This principle is crucial in the formulation of the HyMoLAP model as it ensures the stability of the watershed's geomorphological properties [17]. The HyMoLAP model has been applied in rainfall-runoff simulations and has demonstrated good performance, as highlighted in [14,18,19]. For example, in [19], an adjustment of the model was applied to 20 sub-catchments in Africa and the United States (US), demonstrating on average similar performance to the conceptual Génie Rural à 4 paramètres Journalier (GR4J) [20] model, while exceeding the performance of the light version of the conceptual Hydrologiska Byråns Vattenbalansavdelning (HBV) [21] model. Conversely, the HyMoLAP model offers a significant advantage by simulating river basin discharges. This advantage lies in the incorporation of uncertainties through the associated Stochastic Differential Equations (SDEs). This approach allows the model to account for the inherent random fluctuations in the rainfall-runoff process by utilizing noises that can be either Gaussian or non-Gaussian, as recently demonstrated by [22].

The potential of machine learning techniques for quantifying hydrological processes is widely recognized and continues to demonstrate promising results. It was highlighted in [23] that in addition to hydrological models, the upcoming and promising trend of incorporating machine learning in hydrology can provide the basis for future efforts in supporting operational water management by providing the needed information efficiently and with the required accuracy. Due to their adaptability and capacity to uncover hidden patterns within extensive hydrological and meteorological datasets, machine learning models have demonstrated superior accuracy compared to conventional modeling methods, even in ungauged basins lacking measured runoff data [24]. Machine learning models have seen extensive use in hydrological modeling over the past two decades, demonstrating their potential to enhance predictive capabilities [25–30]. For instance, models such as Support Vector Machine (SVM), Random Forest (RF), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU) have been applied in [31], [32], [33], and [34], respectively, for streamflow simulations. Recent advancements include the hybrid LSTM network with an Ant Lion Optimizer (LSTM-ALO), used in [35] for monthly runoff simulation, and the hybrid Extreme Learning Machine tuned with the Improved Reptile Search Algorithm (ELM-IRSA), which demonstrated superior performance in predicting river flows in [36]. Similarly, the study in [37] investigated a hybrid Relevance Vector Machine tuned with the Dwarf Mongoose Optimization Algorithm (RVM-DMOA), finding that it significantly improved the accuracy of single RVM models in monthly streamflow prediction. In addition to the above machine learning models used in hydrology, other advanced approaches have also been employed for time series prediction. For instance, models such as the Relevance Vector Machine tuned with the Improved Manta-Ray Foraging Optimization (RVM-IMRFO), hybrid Adaptive Neuro-Fuzzy Inference System (ANFIS-WCAMFO), and the LSTM-weighted mean of

vectors optimizer (LSTM-INFO) have demonstrated remarkable performance in studies like [38], [39] and [40].

The best machine learning model depends on the specific context of the study, the characteristics of the data, and the objectives of the analysis, which can vary across various fields, including hydrology. The study of [41] demonstrated that the eXtreme Gradient Boosting (XGBoost) is applicable for streamflow forecasting, and in general, performs better than SVM model. Moreover, a comprehensive study conducted by [27] explored the use of deep learning approaches to predict streamflows, focusing on the application of Recurrent Neural Networks (RNN), LSTM and GRU models across three different streamflow observation stations. The study found that the GRU model delivered the most efficient estimation results for streamflow prediction in these cases, suggesting its potential for applications in other water resource fields. The study in [25] investigated the effectiveness of various machine learning models for predicting daily runoff in the mountainous Sainji catchment in the Himalayas. The models compared include multivariate adaptive regression splines (MARS), wavelet-based artificial neural networks (WANN), and standalone artificial neural networks (ANN). Under the unique hydrological conditions of the region, the findings indicated that both the MARS and WANN models outperformed the ANN model in terms of predictive accuracy. The integration of wavelet transforms, a powerful data preprocessing technique, with machine learning models for rainfall-runoff modeling has gained increasing popularity in recent years [42,43,44]. By decomposing time series data into various frequency components, wavelet transforms facilitate the extraction of critical information that improves model accuracy and reliability.

Hybrid modeling, i.e., the combination of both approaches within one modular hydrological model, can yield improved forecasts [45]. It tries to combine the advantages of both process-oriented and data-oriented modeling approaches [46]. There are different ways to combine rainfall-runoff modeling with machine learning techniques, and error prediction is one of them. For example, in [47], the authors employed two machine learning models to predict the errors in the GR4J hydrologic model, and the results indicated the superiority of the hybrid approach, showing a significant reduction in prediction errors for low flows. In [48], three hybrid models incorporating watershed memory and residual error were developed and evaluated against SWAT and LSTM models, demonstrating superior performance. The author also highlighted that the choice between watershed memory-based or residual error-based hybrid models may depend on the specific study focus. The hybrid models outperformed standalone LSTM model across all watersheds. On the other hand, one of the frequently adopted hybridizations within the scope of rainfall-runoff modeling rests on directing various outputs simulated from the conceptual rainfall-runoff models to machine learning techniques [49]. Using this approach, DeepGR4J—a deep learning hybridization with the conceptual GR4J model—was developed in [5]. The authors considered deep learning models such as Convolutional Neural Networks (CNN) and LSTM, and compare various hybrid models with both conceptual and data-driven models. They also investigated the impact of incorporating streamflow observations from previous time steps into the model inputs. Their study demonstrated that the hybrid models enhanced the accuracy of the stand-alone models, showing greater performance improvements when streamflow observations from previous time steps were included. Similarly, in [4], the outputs of the conceptual CemaNeige Génie Rural à 6 paramètres Journalier (GR6J), which includes an integrated snow module at a daily scale, were used as input data for Wavelet-based Support Vector Regression (SVR) and Wavelet-based Multivariate Adaptive Regression Spline (WMARS) to enhance the performance of both canonical conceptual and machine learning models.

The hybridization approach that incorporates hydrological model outputs as inputs for machine learning models has demonstrated superior performance and continues to gain popularity. While these hydrological models are deterministic, this hybridization approach can be extended to integrate conceptual or physically-based stochastic rainfall-

Table 1

Statistics for the daily precipitation, potential evapotranspiration (PET) and discharge data over the selected period 2000-2005.

Data	Statistics						
	Minimum	Mean	Standard deviation	Median	Maximum	Skewness	Kurtosis
Precipitation [mm]	0.000	3.029	4.536	0.809	31.471	2.129	5.279
PET [mm]	2.112	3.988	0.744	4.046	6.189	-0.101	-0.443
Discharge [m ³ /s]	6.376	170.311	253.375	28.720	971.500	1.557	1.107

runoff models with machine learning techniques by using the statistical properties of discharge probability distributions simulated by the hydrological models as input data for machine learning frameworks. This could allow for the advantage of exploring information on uncertainties to improve discharge predictions.

The aim of this study is to enhance rainfall-runoff modeling by developing a novel hybrid approach, which combines stochastic hydrological models with machine learning methods, while maintaining better interpretability compared to standalone machine learning frameworks. This was achieved by integrating a stochastic version of the Hydrological Model based on the Least Action Principle (Sto. HyMoLAP) with machine learning models, such as Wavelet-based eXtreme Gradient Boosting (WXGBoost) and Wavelet-based Gated Recurrent Unit (WGRU). The stochastic rainfall-runoff model Sto. HyMoLAP, used and recently developed in [22], is a Langevin-type SDE. Non-Gaussian Lévy stable noise is used to represent uncertainties in the rainfall-runoff system. Baseline models, such as XGBoost and GRU, are employed rather than advanced or optimized approaches like LSTM-ALO and ELM-IRSA mentioned above, to demonstrate that the proposed hybridization technique can deliver effective performance even with simpler machine learning frameworks. The Ouémé at Bonou river basin was used exclusively for the application of the methods in this study. The following objectives are outlined:

- Developing hybrid models that integrate the statistical properties of daily probabilistic discharge derived from stochastic rainfall-runoff models into the machine learning process;
- Assessing the performance of the hybrid models compared to the standalone rainfall-runoff and machine learning models in the Ouémé at Bonou river basin;
- Analyzing how the ranges of uncertainties associated with daily discharge influence the performance of the hybrid approaches.

The model's performance assessment includes an evaluation of overall performance using various metrics, followed by an error analysis of the models across different discharge classes. The paper is organized as follows: section 2 presents the data used, the different standalone models, and the formulation of the hybrid models, followed by the model performance metrics. In section 3, the various results are presented and discussed. The paper ends with a general conclusion in section 4.

2. Methodology

2.1. Data used

This paper develops a new hybridization technique for advanced rainfall-runoff modeling, applicable to any river basin. The approach was validated using the Ouémé at Bonou river basin as the study area. Indeed, the Ouémé river is the largest river in the Republic of Benin, originating from the Tanéka classified forest (Atacora). Benin shares the Guinea Coast of West Africa (between 6°25' and 12°30' North latitude and between 0°45' and 4°East longitude) and is bordered to the west by Togo, to the east by Nigeria, and to the north by Niger and Burkina Faso [18]. The Ouémé at Bonou river basin is a sub-catchment of the Ouémé river, covering a surface area of 49,256 km².

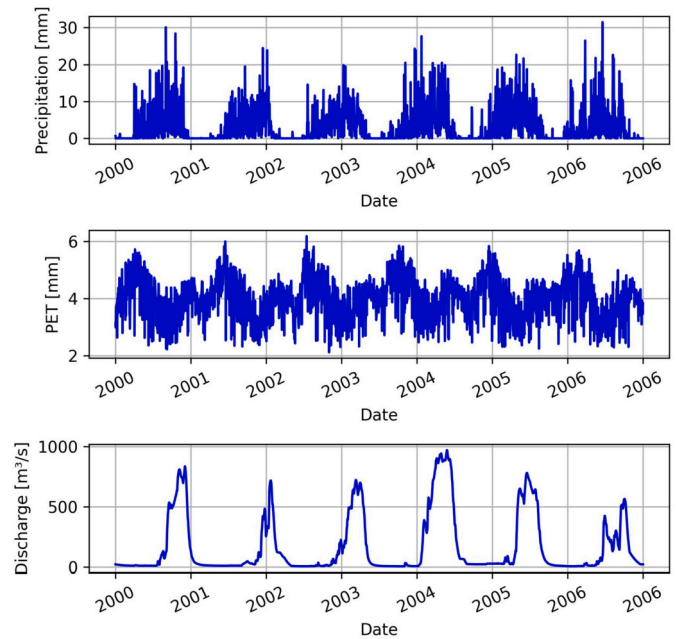


Fig. 1. Daily time series plot of precipitation, potential evapotranspiration (PET), and discharge in the Ouémé at Bonou river basin.

This study utilized daily discharge, precipitation, and potential evapotranspiration data, as shown in Fig. 1. Due to missing data in the received series, the period 2000-2005 was selected. The daily precipitation and potential evapotranspiration data were provided by Météo-Benin, while the discharge data were obtained from the National Directorate of Water (DG-Eau), Benin. Table 1 presents basic statistics for the three data over the selected period. The precipitation data reveals a highly variable pattern, with frequent dry days and occasional heavy rainfall, as indicated by the positive skewness and high kurtosis. In contrast, potential evapotranspiration (PET) shows low variability and a nearly symmetrical distribution, suggesting stable atmospheric conditions. River discharge exhibits significant variability, with a right-skewed distribution and elevated kurtosis, reflecting that most days have moderate flow, but there are sporadic instances of high discharge, likely corresponding to flood events triggered by heavy rain. As in [22], the datasets were utilized such that two-thirds (2000-2003, approximately 66.66%) of the data were allocated for calibration, while one-third (2004-2005, approximately 33.33%) was reserved for validation. This approach aligns with common practices in model evaluation, ensuring that the model is adequately trained on a substantial portion of the data while maintaining an independent set for assessing its predictive performance.

2.2. Physically-based stochastic hydrological model

The physically-based stochastic HyMoLAP (Sto. HyMoLAP) model utilized is the formulation introduced in [22]. This model is a Lévy-driven SDE model of Langevin type, formulated as a stochastic version of the Hydrological Model based on the Least Action Principle (HyMoLAP) [13-15,18]. In fact, the HyMoLAP model is grounded in fundamental

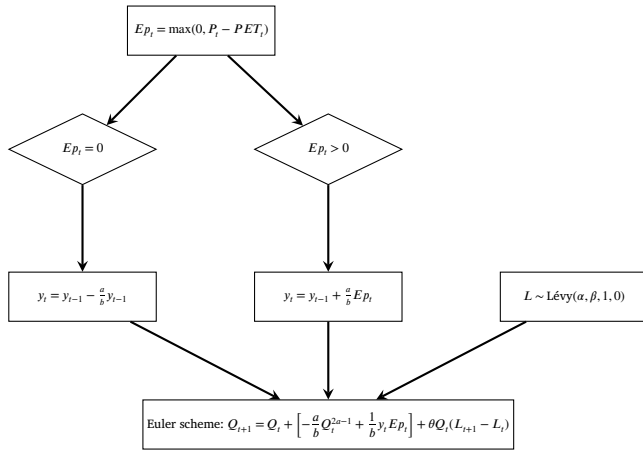


Fig. 2. Schematic representation of the simulation process for the stochastic model (Sto. HyMoLAP).

physical principles, specifically aimed at minimizing the energy within the system. The model takes precipitation (P) and potential evapotranspiration (PET) as inputs. The deterministic form (Det. HyMoLAP) includes two parameters: a and b , which represent the nonlinearity parameter and the macroscopic parameter, respectively, describing emergent properties related to the geomorphology and pedology of the river basin [15]. The stochastic model includes additional parameters and is given by:

$$dQ(t) = \left[-\frac{a}{b} Q(t)^{2a-1} + \frac{1}{b} \psi(q, t) \right] dt + \theta * Q(t) dL(t), \quad (1)$$

where $\psi(q, t)$ is a composite forcing function, that we define in the form $\psi(q, t) = y(t)Ep(t)$, as in [22]. Using Euler-Maruyama discretization scheme [50], the discrete form of the model is

$$Q_{t+1} = Q_t + \left[-\frac{a}{b} Q_t^{2a-1} + \frac{1}{b} y_t Ep_t \right] + \theta Q_t (L_{t+1} - L_t), \quad (2)$$

with,

$$y_t = \begin{cases} y_{t-1} - \frac{a}{b} y_{t-1}, & \text{if } Ep_t = 0 \\ y_{t-1} + \frac{a}{b} Ep_t, & \text{if } Ep_t > 0 \end{cases}, \quad (3)$$

where Ep_t represents the effective precipitation, calculated as $Ep_t = P_t - PET_t$. If $P_t < PET_t$, then Ep_t is set to 0. θ ($\theta \in \mathbb{R}^*$) represents a constant parameter, and L_t denotes a Lévy stable motion with the following characteristics: a stability index $\alpha \in (0, 2]$, an asymmetry parameter $\beta \in [-1, 1]$, a scale parameter (set to 1), and a location parameter (set to 0).

The model (1) was calibrated using the algorithmic method developed in [22]. This method is based on Monte Carlo simulation of SDE [51], Euler-Maruyama discretization scheme, and Nash-Sutcliffe Efficiency (NSE) [52] criterion. Considering $y(0) = P(0) - PET(0)$, the simulation process with the model is illustrated in Fig. 2. In certain river basin studies, discharge forecasts using the HyMoLAP model may be significantly influenced by specific basin data characteristics or by the Lévy noise (in the stochastic model). This influence can occasionally produce large simulated values, leading to negative flows, which are typically expected to be positive or close to zero. As example, an adjustment like $Q_{t+1} = \max(Q_{t+1}, 0)$ can be applied to reflect basin characteristics that prevent negative streamflows, thus ensuring physically realistic values.

2.3. Machine learning methods

2.3.1. Wavelet transformation

Wavelets are mathematical functions that use time-scale representations to analyze time series that may contain nonstationarities [25]. Their ability to handle multiscale data and remove noise while preserv-

ing relevant information has made them a valuable tool in enhancing data quality [53]. Various types of wavelet transformation exist, including the Discrete Wavelet Transform (DWT), commonly used for analyzing discrete data in applications like signal processing and data compression, the Continuous Wavelet Transform (CWT), which is applied to continuous signals and is particularly useful for time-frequency analysis, and the Stationary Wavelet Transform (SWT), a shift-invariant variant of the DWT used for time-series data without downsampling. In hydrology, Wavelet transformation has been applied for various purposes, including trend detection, handling nonstationary data, and data denoising [54]. The DWT is the most widely used wavelet transform, valued for its computational efficiency and suitability for inherently discrete digital data. Additionally, the DWT provides information about the signal at specific scales and has been utilized in rainfall-runoff modeling studies [54,55]. Another advantage of using the DWT is that it allows for the implementation of digital filtering effects [25]. High and low pass filters can be readily implemented, resulting in detailed coefficients and approximation series [56]. The coefficients of the DWT are obtained according to the level of decomposition, the time translation factor, a fixed dilation step, and the location parameter [57]. The DWT is characterized by its use of various mother wavelets, representing different families of wavelets, which define the shape and properties of the wavelet functions, allowing for tailored analysis suited to different types of data characteristics.

In this study, the DWT was employed to enhance the performance of various standalone machine learning and hybrid models. The simulations showed that omitting this transformation resulted in diminished performance. Consequently, different mother wavelets of the DWT, specifically Daubechies 1 (db1), Daubechies 2 (db2), Coiflet 1 (coif1), Coiflet 2 (coif2), and Symlet 2 (sym2), were investigated for each model. By evaluating these mother wavelets across various decomposition levels, we identified the optimal combination of mother wavelet and decomposition level for the DWT that delivered the best performance for each model within the applicable context of this research.

2.3.2. Extreme Gradient Boosting (XGBoost)

XGBoost, developed in 2016 by [58], is one of the new generation machine learning algorithms, which increases the model's overall accuracy by preventing the problem of overfitting in the process of training the algorithm [59]. It is an ensemble algorithm that incorporates gradient boosting methods [60]. This advanced implementation of gradient boosting is designed for speed and performance. It employs decision trees as a base learner and builds multiple weak learners in an iterative process [61]. The algorithm iteratively adds new models that correct the errors of existing ones, leading to improved performance. XGBoost has gained popularity due to its efficiency, flexibility, and accuracy in predictive modeling tasks, especially in machine learning competitions like those hosted by Kaggle [62]. One of XGBoost's strengths is its ability to handle missing values effectively by assigning default directions during the split process. This optimization allows the algorithm to determine the best split for both known and missing values, leading to a more robust model that retains valuable information without extensive data preprocessing.

The objective function to be minimized is a combination of a loss function and a regularization term to penalize complexity. It is defined as [63]:

$$\text{Obj}(\theta) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K R(f_k) + C, \quad (4)$$

where,

- l is a differentiable convex loss function that measures the difference between the prediction $\hat{y}_i$ and the target y_i , (e.g., Mean Squared Error or Logistic Loss);
- C is a constant, that is independent of the parameters being optimized;

- $R(f_k)$ denotes the regularization term at the k iteration time to control overfitting, typically defined as:

$$R(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2, \quad (5)$$

where γ denotes complexity of leaves, T is the number of leaves, λ is the penalty parameter and w_j is the output of each leaf node. Leaves represent the expected categories based on classification criteria, whereas the leaf node denotes the tree node which cannot be divided [64].

The XGBoost model improves its predictions by minimizing the objective function at each iteration. To do this, the algorithm uses approximations of the objective function based on the gradient (first derivative) and the Hessian (second derivative):

$$g_i = \frac{\partial l(y_i, \hat{y}_i)}{\partial \hat{y}_i}, \quad (6)$$

$$h_i = \frac{\partial^2 l(y_i, \hat{y}_i)}{\partial \hat{y}_i^2}. \quad (7)$$

The gradient g measures the direction and magnitude in which the loss function can be reduced, while the Hessian h captures the local curvature of the function, allowing for more accurate weight updates. These two terms are essential for adjusting the weights in the decision trees, thereby improving the accuracy of predictions with each iteration. The XGBoost approach enables the development of a high-performing model while effectively mitigating the risks of overfitting.

2.3.3. Gated Recurrent Unit (GRU)

Recurrent Neural Networks (RNNs) are able to effectively capture time dependencies in time series data by maintain a memory (state) of previous inputs while processing new inputs [65]. Their topology is in the form of layers of connected nodes, and it requires two main parameters: the number of layers and the number of nodes [66]. Models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are two types of RNNs that have demonstrated their enormous effectiveness for streamflow point prediction [27,67–70]. They have become a popular choice for recurrent neural networks due to their effectiveness and efficiency [53]. The distinctive capability of the LSTM and GRU is learning non linear long term dependency structures from data. The GRU, introduced by [71], is a streamlined variant of the LSTM network, first proposed by [72], designed to simplify the internal structure of the model while retaining its ability to capture both short-term and long-term dependencies in sequential data. By eliminating the cell state and merging the forget and input gates into a single update gate, the GRU reduces the computational complexity found in LSTMs, leading to faster convergence and fewer parameters to train.

Given a time series $X_1, \dots, X_t$, where $X_i = (X_{1,i}, \dots, X_{n,i})^T$ for $i = 1, \dots, t$, the GRU manages its memory with two gates: the reset gate and the update gate. The GRU's computational process involves fewer operations than the LSTM. Mathematically, as described by [71], the GRU equations are as follows:

$$\begin{aligned} z_{(t)} &= \sigma(W_{hz} \cdot h_{(t-1)} + W_{xz} \cdot X_{(t)} + b_z) \\ r_{(t)} &= \sigma(W_{hr} \cdot h_{(t-1)} + W_{xr} \cdot X_{(t)} + b_r) \\ \hat{h}_{(t)} &= \tanh(W_{hh} \cdot (r_{(t)} \odot h_{(t-1)}) + W_{xh} \cdot X_{(t)} + b_h) \\ h_{(t)} &= (1 - z_{(t)}) \odot h_{(t-1)} + z_{(t)} \odot \hat{h}_{(t)} \end{aligned} \quad (8)$$

In these equations, W and b are respectively the weight matrices and bias vectors that need to be learned during training, σ is the activation function, and $\odot$ denotes the Hadamard (element-wise) product.

Moreover, at each time step t ,

- $z_{(t)}$: **Update gate**—determines how much of the previous hidden state $h_{(t-1)}$ should be carried forward. If $z_{(t)}$ is close to 1, most of the previous state is preserved, allowing the GRU to remember information over long periods. Conversely, if $z_{(t)}$ is close to 0, the model focuses on the current input;
- $r_{(t)}$: **Reset gate**—controls how much of the previous hidden state should influence the candidate hidden state. A low value of $r_{(t)}$ results in the model forgetting the past and focusing primarily on the new input, while a high value incorporates more of the past information;
- $\hat{h}_{(t)}$: **Candidate hidden state**—a temporary computation based on the reset gate and the current input $X_{(t)}$, scaled by the reset gate $r_{(t)}$;
- $h_{(t)}$: **Final hidden state**—a combination of the previous hidden state and the candidate hidden state, weighted by the update gate $z_{(t)}$.

The GRU's reset and update gates serve as efficient mechanisms for retaining essential information while forgetting irrelevant data over time. In scenarios where both LSTM and GRU models perform similarly, the GRU's computational efficiency and reduced memory usage often make it the more attractive option.

2.4. Formulation of the proposed hybrid models

The hybrid models in this study represent an integration of the physically-based stochastic hydrological model (1) with machine learning. This approach relies on incorporating the simulation results from the stochastic hydrological model as inputs for the machine learning models. To this end, we focused solely on the daily simulated statistical properties of the discharge probability distribution. These statistical properties are first estimated from the simulations of the model (1), considering a large number of simulations using the Monte Carlo approach [51]. This method ensures a robust exploration of the uncertainty in the model by generating numerous trajectories, from which we can estimate statistical characteristics of the daily probability density functions. Among all the metrics that can be computed, the mean ($\bar{Q}_{sto}$) and different quantile trajectories were considered in this study. The input sets used are $(\bar{Q}_{sto})$ and $(\bar{Q}_{sto}, q_u, q_v, q_v - q_u)$, where q_u is the quantile at position u and $q_v - q_u$ represents the quantile range or the interval width between the quantiles at positions u and v . For example, $q_{0.984} - q_{0.016}$ refers to the range between the value at the 98.4th percentile and that at the 1.6th percentile, indicating the spread of daily simulated values through Monte Carlo approach, that lie between these two quantiles. Additionally, the hybrid models and the standalone machine learning models considered various lag times, such as t , $(t - 1)$ and $(t - 2)$, to predict the discharge at $(t + 1)$.

The Wavelet transformation was employed prior to the standalone machine learning and hybrid models. This transformation decomposes the inputs into low-frequency components, reducing noise while retaining important structural information. Once the data had been transformed, in the case of XGBoost model, the features were normalized using the *StandardScaler*, which guarantees a mean of zero and a standard deviation of one for each feature. This ensures optimal convergence of the XGBoost model, which is sensitive to the scales of the variables. In contrast, for the GRU model, the *MinMaxScaler* was applied. This scaler transforms the data to values within the range of 0 and 1, thereby enhancing the stability and learning speed of the GRU model. The schematic representation of the proposed hybrid models is shown in Fig. 3.

2.5. Model performance metrics

Model performance metrics are essential tools for assessing the quality of predictive models in various fields, such as hydrology or artificial

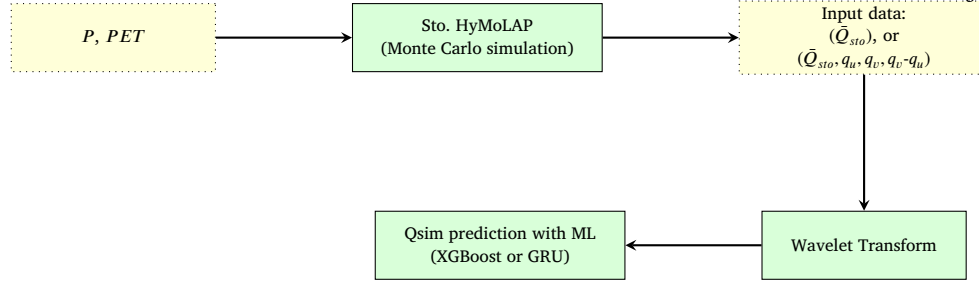


Fig. 3. Structure of the Sto. HyMoLAP+Machine Learning hybrid models.

Table 2
Model evaluation metrics in this study.

Metric	Mathematical formula	Range	Optimal value
NSE [52]	$1 - \frac{\sum_{i=1}^N (Q_{obs,i} - Q_{sim,i})^2}{\sum_{i=1}^N (Q_{obs,i} - \bar{Q}_{obs})^2}$	$(-\infty, 1]$	1
R ² [78]	$\left[\frac{\sum_{i=1}^N (Q_{obs,i} - \bar{Q}_{obs})(Q_{sim,i} - \bar{Q}_{sim})}{\sqrt{\sum_{i=1}^N (Q_{obs,i} - \bar{Q}_{obs})^2} \sqrt{\sum_{i=1}^N (Q_{sim,i} - \bar{Q}_{sim})^2}} \right]^2$	$[0, 1]$	1
RMSE [79]	$\sqrt{\frac{1}{N} \sum_{i=1}^N (Q_{obs,i} - Q_{sim,i})^2}$ (m^3/s)	$[0, +\infty)$	0
MAE [75]	$\frac{1}{N} \sum_{i=1}^N Q_{obs,i} - Q_{sim,i} $ (m^3/s)	$[0, +\infty)$	0
CA [75]	$0.33(RMSE + MAE + (1 - R^2))$	$[0, +\infty)$	0

intelligence. They are used to estimate how well a model reproduces observed phenomena and predicts future behavior. This study employed commonly used metrics, including the Nash-Sutcliffe coefficient (NSE), which assesses the correspondence between observed and simulated values; the coefficient of determination (R^2), which measures the proportion of variance explained by the model; the root mean square error (RMSE), which quantifies the average magnitude of errors between predicted and actual values [73]; and the Mean Absolute Error (MAE), which indicates the degree of error in the predictions, without considering the direction [74]. Additionally, the Combined Accuracy (CA), is employed as in [75,76]. The CA is a comprehensive metric that integrates RMSE, MAE, and an inverse evaluation of model performance based on R^2 . In fact, the NSE remains the primary criterion for assessing overall performance in hydrology, given its sensitivity to high values that characterize flood situations. Based on [77], an NSE value between 0.8 and 1 indicates very good performance, values from 0.7 to 0.8 are considered good, between 0.5 and 0.7 is deemed satisfactory, while values below 0.5 reflect unsatisfactory performance. Simply, an NSE value close to 1 indicates that the model reproduces observations well. Similarly, an R^2 value close to 1 means that the model explains a large proportion of the variance in the observations. On the other hand, a small RMSE or MAE indicates that the model's predictions are close to the observed values. The CA metric accounts for large errors through RMSE and MAE while rewarding a high proportion of explained variance using R^2 . Its optimal value is 0, with smaller values indicating better performance. Table 2 presents the formulas, ranges, and optimal values for these three metrics. By using a combination of different metrics, one can evaluate the overall performance of a model, fine-tune its parameters, and enhance its predictive capabilities for practical applications.

3. Results and discussion

The project is developed using Python version 3.10.12 and integrates important libraries such as NumPy, Pandas, scikit-learn, Keras, as well as scipy.stats and pywt.

Table 3
Calibrated parameters of the Sto. HyMoLAP model.

Models	Calibrated parameters				
	a	b	θ	α	β
Det. HyMoLAP	0.8158	2.9242	-	-	-
Sto. HyMoLAP	0.8136	2.9416	0.0035	1.3	-0.8

3.1. Standalone physically-based and machine learning models

For $\theta = 0$, the model (1) becomes deterministic and was calibrated by adapting the same calibration method. Table 3 presents the calibrated parameters for both the deterministic and stochastic HyMoLAP models across the Ouémé at Bonou river basin. The same calibrated parameter results of the stochastic model in [22] were used, as shown in Table 3. Both models were evaluated for validation. In the case of the stochastic model, the Monte Carlo approach was applied with a total of $N_{traj} = 300,000$ trajectories to generate independent daily discharge paths. The results do not show a significant difference when analyzing the average trajectory derived from the 300,000 simulated trajectories compared to the 5,000 trajectories considered in [22]. However, it is clear that $N_{traj} = 300,000$ clearly offers a better representation of the system's stochasticity. When considering the average trajectory of the stochastic model with $N_{traj} = 300,000$, both models produce similar results in terms of NSE, R^2 , RMSE and CA criteria as presented by Table 6. The average trajectory closely matches the deterministic model's simulation, reinforcing the common assumption in stochastic modeling that system uncertainties (or noise) introduced by the stochastic model do not systematically bias the results in one direction. Instead, their impact tends to average out over time, converging toward the deterministic solution.

In the case of the WXGBoost model, as well as for WXGBoost-based hybrid models, we used a *RandomizedSearchCV* to optimize the hyperparameters of the XGBoost model, exploring 50 combinations of hyperparameters defined in a grid, with 5-fold cross-validation (*KFold*), where shuffling was enabled (*shuffle = True*) to ensure a random shuffling of the data at each epoch, thereby improving the generalization of the models, and using the *neg_mean_squared_error* metric as the evaluation criterion to minimize the mean squared error. As is commonly done, during training, we defined sets of values, and the optimal values obtained after applying *RandomizedSearchCV*, as well as the optimal wavelet family and decomposition level used for training the WXGBoost model, are presented in Table 4. The optimal values are underlined within the tested parameter sets. Moreover, the performance of the model is provided in Table 6. It can be seen that the model demonstrates lower performance compared to physically-based models; however, its performance improves as the lag time increases to $(t - 2)$.

The WGRU model, as well as WGRU-based hybrid models, have been uniformly configured for certain training parameters. Specifically, each model uses the *relu* activation function in its GRU layers, allowing complex relationships to be captured through the introduction of non-linearity. The *Adam* optimizer was selected for its efficiency. In addition, the loss function used is the *mean squared error (mse)*, and finally, the *shuffle* parameter is activated (*shuffle = True*). The other obtained train-

Table 4
Characteristics of WXGBoost models yielding the best performance.

Models	DWT		Hyperparameters					
	type	level	$n_{estimators}$	max_depth	$learning_rate$	$colsample_bytree$	reg_alpha	reg_lambda
WXGBoost	sym2	6	[50, 100, <u>150</u>]	[2, 3, <u>5</u>]	[0.01, <u>0.05</u>]	[0.5, <u>0.7</u> , 0.9]	[0.1, 0.3, <u>0.5</u>]	[0.1, 0.3, <u>0.5</u>]
Sto. HyMoLAP-WXGBoost ₁	coif1	4	[50, <u>100</u> , 150]	[2, 3, 5]	[0.01, <u>0.05</u>]	[0.5, <u>0.7</u> , 0.9]	[<u>0.1</u> , 0.3, 0.5]	[0.1, <u>0.3</u> , 0.5]
Sto. HyMoLAP-WXGBoost ₂	coif1	4	[50, <u>100</u> , 150]	[2, 3, 5]	[0.01, <u>0.05</u>]	[0.5, <u>0.7</u> , 0.9]	[0.1, 0.3, <u>0.5</u>]	[<u>0.1</u> , 0.3, 0.5]

Table 5
Characteristics of WGRU models yielding the best performance.

Models	DWT		Hyperparameters			
	type	level	$nb_neurons$	$epochs$	$batch_size$	nbr_GRU_layers
WGRU	coif1	4	85	200	512	2
Sto. HyMoLAP-WGRU ₁	db2	3	95	200	128	1
Sto. HyMoLAP-WGRU ₂	db2	3	95	700	2048	3

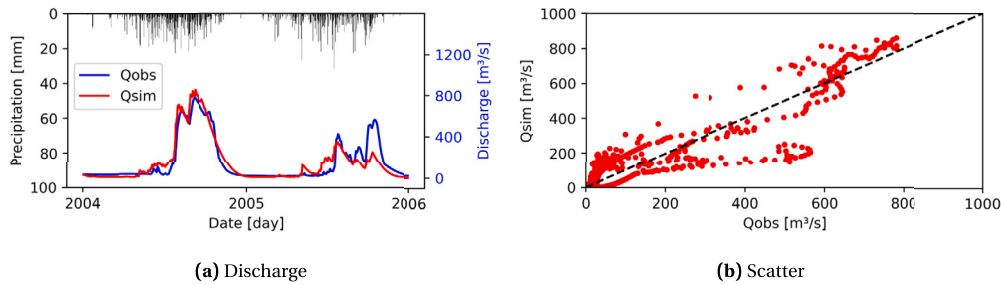


Fig. 4. Comparison of observed and simulated discharge, derived from the HyMoLAP model in the validation phase. Qobs is the observed discharge, while Qsim denotes the simulated discharge.

ing hyperparameters, as well as the wavelet family and decomposition level used for WGRU, are shown in Table 4. These optimal characteristics were obtained after the hyperparameter tuning process. Based on the evaluation criteria displayed in Table 6, it can be seen that the standalone WGRU model provides the poorest performance in this study.

Overall, all the aforementioned standalone models were trained to simulate daily discharge using only precipitation (P) and potential evapotranspiration (PET) data. The standalone physically-based HyMoLAP models demonstrate strong performance when compared to the standalone machine learning models, as displayed in Table 6. This finding highlights a limitation observed in this study regarding machine learning models, as they struggled to predict discharge using only observed precipitation and potential evapotranspiration data. Similar results were obtained in the studies by [80,5,4], which demonstrated that with these two inputs alone, machine learning models provide less accurate results compared to the hydrological models employed. This limitation is likely due to the fact that machine learning models rely heavily on patterns within the data and may struggle to capture the underlying physical processes governing hydrological systems when only these two input variables are used. In contrast, hydrological models such as HyMoLAP incorporate physical principles and prior knowledge of the system, allowing them to simulate complex interactions more effectively. The subsection 3.2 will delve deeper into the performance of hybrid models, which integrate the two machine learning models with the physically-based stochastic approach to leverage the strengths of both methodologies and enhance the results obtained from the standalone models. To this end, we will consider the results of the physically-based models, as displayed in Fig. 4 and Table 6, and presented in [22], as the baseline for comparison.

3.2. Hybrid models

The hybrid models were developed considering two different main scenarios: one where only the mean discharge, simulated by the

physically-based stochastic model, is taken into account, and another where various quantile trajectories and their ranges are included. The input sets used are $(\bar{Q}_{sto})$ and $(\bar{Q}_{sto}, q_u, q_v, q_w, q_{u-}, q_{u+})$, respectively. In both scenario cases, the features of the hybrid models are presented in Tables 4 and 5. Table 6 presents NSE, R^2 , RMSE and CA performance results for training and test sets in daily prediction tasks for the Ouémé at Bonou river basin dataset. We compare four variations of the hybrid models: Sto. HyMoLAP-WXGBoost₁, Sto. HyMoLAP-WGRU₁, Sto. HyMoLAP-WXGBoost₂, and Sto. HyMoLAP-WGRU₂, against the standalone physically-based (HyMoLAP) and machine learning (WXGBoost and WGRU) models. Out of the various models analyzed, the WGRU-based hybrid model, incorporating quantile trajectory information (Sto. HyMoLAP-WGRU₂), demonstrates the best overall performance across all considered metrics, followed by the Sto. HyMoLAP-WXGBoost₂ hybrid model. However, when using $(\bar{Q}_{sto})$ as input (first scenario), only the Sto. HyMoLAP-WGRU₁ model improves the accuracy of the standalone HyMoLAP model.

Subplots 5a to 5d of Fig. 5 display the hydrographs with the predicted discharges and the associated scatter plots from the Sto. HyMoLAP-WXGBoost₁ and Sto. HyMoLAP-WGRU₁ hybrid models, corresponding to the first scenario. It is observed that incorporating WGRU with the physically-based model was beneficial in this context. In this study, with $(\bar{Q}_{sto})$ as only input data for the hybrid approach, the Sto. HyMoLAP-WGRU₁ hybrid model better handled the nonlinear relationships inherent in the data, leading to more accurate predictions. However, it is evident that the hybrid model does not improve the prediction of the high observed discharge in the second year. This similarity is evident even in the corresponding scatter plot, which resembles that of the standalone HyMoLAP model, as shown in Fig. 4. We emphasize that this hybridization scenario can be regarded as the integration of the deterministic HyMoLAP model with machine learning, as a high number of simulated trajectories by HyMoLAP causes $(\bar{Q}_{sto})$ to become indistinguishable from the simulation result of the deterministic model presented in Fig. 4.

Table 6
Performance of the standalone and hybrid models.

Models	Input data	Model evaluation metrics							
		Calibration (2000-2003)				Validation (2004-2005)			
		NSE	R ²	RMSE (m ³ /s)	CA	NSE	R ²	RMSE (m ³ /s)	CA
Det. HyMoLAP	$(P, PET)_t$	0.881	0.881	92.924	48.837	0.835	0.843	87.677	47.527
Sto. HyMoLAP	$(P, PET)_t$	0.886	0.885	91.557	47.934	0.836	0.842	87.325	47.101
WXGBoost	$(P, PET)_t$	0.452	0.458	199.736	113.554	0.433	0.459	162.693	96.353
	$(P, PET)_{t-1,t}$	0.907	0.923	82.474	42.598	0.596	0.600	137.324	75.266
	$(P, PET)_{t-2,t-1,t}$	0.928	0.942	72.511	35.810	0.691	0.692	120.137	66.046
WGRU	$(P, PET)_t$	0.453	0.457	199.501	112.193	0.280	0.331	183.307	107.782
	$(P, PET)_{t-1,t}$	0.516	0.518	187.470	104.924	0.267	0.354	185.041	108.666
	$(P, PET)_{t-2,t-1,t}$	0.531	0.532	184.686	103.012	0.345	0.400	174.964	101.389
Sto. HyMoLAP-WXGBoost ₁	$(\hat{Q}_{sim})_t$	0.903	0.903	84.156	42.913	0.817	0.832	92.353	47.277
	$(\hat{Q}_{sim})_{t-1,t}$	0.903	0.903	84.203	42.907	0.815	0.833	92.838	47.350
	$(\hat{Q}_{sim})_{t-2,t-1,t}$	0.904	0.905	83.421	42.486	0.804	0.827	95.607	49.265
Sto. HyMoLAP-WGRU ₁	$(\hat{Q}_{sim})_t$	0.883	0.890	92.198	49.066	0.851	0.853	83.312	44.483
	$(\hat{Q}_{sim})_{t-1,t}$	0.886	0.892	91.203	47.575	0.854	0.854	82.703	43.717
	$(\hat{Q}_{sim})_{t-2,t-1,t}$	0.889	0.891	89.841	46.922	0.859	0.863	81.032	42.072
Sto. HyMoLAP-WXGBoost ₂	$(\hat{Q}_{sim}, q_{0.047}, q_{0.953}, q_{0.953} \sim q_{0.047})_t$	0.986	0.987	31.690	13.174	0.860	0.879	80.855	42.670
	$(\hat{Q}_{sim}, q_{0.028}, q_{0.972}, q_{0.972} \sim q_{0.028})_{t-1,t}$	0.989	0.990	27.304	10.961	0.870	0.890	78.035	41.479
	$(\hat{Q}_{sim}, q_{0.027}, q_{0.973}, q_{0.973} \sim q_{0.027})_{t-2,t-1,t}$	0.994	0.994	20.629	8.134	0.881	0.901	74.715	40.604
Sto. HyMoLAP-WGRU ₂	$(\hat{Q}_{sim}, q_{0.08}, q_{0.92}, q_{0.92} \sim q_{0.08})_t$	0.943	0.943	64.368	31.692	0.894	0.907	70.306	38.311
	$(\hat{Q}_{sim}, q_{0.08}, q_{0.92}, q_{0.92} \sim q_{0.08})_{t-1,t}$	0.944	0.945	63.609	31.561	0.896	0.905	69.223	38.167
	$(\hat{Q}_{sim}, q_{0.10}, q_{0.90}, q_{0.90} \sim q_{0.10})_{t-2,t-1,t}$	0.948	0.948	61.416	30.692	0.896	0.909	69.595	38.110

Subplots 5e to 5h of Fig. 5 show the hydrographs with the predicted discharges and the associated scatter plots from the Sto. HyMoLAP-WXGBoost₂ and Sto. HyMoLAP-WGRU₂ hybrid models, corresponding to the second scenario. By incorporating quantile trajectories and their ranges as additional inputs, both hybrid models exhibit improved performance. Specifically, the Sto. HyMoLAP-WXGBoost₂ achieves a 5.5% improvement, while the Sto. HyMoLAP-WGRU₂ yields a 7.30% improvement in terms of NSE, over the standalone HyMoLAP model. A key observation is the enhanced accuracy of the high discharges in the second year, a task that the standalone HyMoLAP and the hybrid models in the first scenario, did not achieve. Furthermore, when comparing the scatter plots—especially for the Sto. HyMoLAP-WGRU₂ model—it is evident that the data points are less dispersed around the line $y = x$ compared to the other cases. This indicates a more accurate prediction. Additionally, we emphasize that in this scenario, the WGRU model effectively managed the nonlinear relationships inherent in the chosen input data.

The performance of the different standalone and hybrid models has been further examined using the violin plots [81] and Taylor diagram [82]. The density and box plots of the observed and simulated runoff (Fig. 6) show that the standalone machine learning models, WXGBoost and WGRU, poorly match the observed discharges compared to the other models. This discrepancy can be further explained by the fact that, as observed in the figure, these models produced many negative flows. On the other hand, the HyMoLAP and hybrid models show similarities in terms of distribution; however, key differences and detailed analyses can be observed to distinguish them. Most of the observed discharges are concentrated in the lower part of the distribution. Regarding other statistics, such as minimum and median values, it is evident that Sto. HyMoLAP-WXGBoost₂ and Sto. HyMoLAP-WGRU₂ match the observed runoff better than HyMoLAP and the two other hybrid models. Furthermore, it can be observed that HyMoLAP, Sto. HyMoLAP-WXGBoost₁, and Sto. HyMoLAP-WGRU₁ underestimate the maximum values, which explains their inability to simulate the high discharges observed during the second year of the validation phase. Between the two hybrid models, Sto. HyMoLAP-WXGBoost₂ and Sto. HyMoLAP-WGRU₂, the former tends to produce more high values, as shown by the distributions. This may explain why Sto. HyMoLAP-WXGBoost₂ is less consistent than Sto. HyMoLAP-WGRU₂ in overall accuracy evaluations. The Sto. HyMoLAP-

WGRU₂ hybrid model effectively addresses the underestimation of both minimum and maximum discharges observed in the HyMoLAP model.

The Taylor diagram, shown in Fig. 7, illustrates the performance of various models for runoff prediction. It highlights the superior performance of the Sto. HyMoLAP-WGRU₂ and Sto. HyMoLAP-WXGBoost₂ hybrid models. The diagram also reveals the comparatively unsatisfactory performance of the standalone machine learning models, WGRU and WXGBoost, when compared to the physically-based and hybrid models. Notably, when comparing the Sto. HyMoLAP-WGRU₂ and Sto. HyMoLAP-WXGBoost₂ hybrid models, the Sto. HyMoLAP-WGRU₂ demonstrates greater consistency in terms of RMSE, correlation coefficient (R), and standard deviation. For HyMoLAP, Sto. HyMoLAP-WGRU₁ and Sto. HyMoLAP-WXGBoost₁ models, despite their less performance compared to other hybrid models based on RMSE and R values, their standard deviations align more closely with the observed runoff. Among these, the HyMoLAP model shows a standard deviation that is most similar to the observed data. When considering the simulated mean trajectory as input data (first scenario) for the hybrid models, only the Sto. HyMoLAP-WGRU₁ demonstrates improvement, showing a similar standard deviation with smaller RMSE and higher R compared to Sto. HyMoLAP-WXGBoost₁. Across both scenarios (two different input data cases), the WGRU model consistently outperforms WXGBoost within the hybrid frameworks. By incorporating the quantile information simulated by the HyMoLAP model as input data, the hybrid models improve overall performance. However, they are less consistent in terms of standard deviation. The analysis of the Taylor diagram further underscores the strengths of each model and their respective advantages, confirming the superior overall performance of the Sto. HyMoLAP-WGRU₂ hybrid model. In conclusion, the performance of the proposed hybridization technique depends on the choice of appropriate input data simulated by the hydrological model and the machine learning model employed. This relationship should be further investigated, depending on the specific study case, to determine the optimal combination that can enhance hydrological model simulations.

In both scenarios of input data, the results demonstrate that integrating the WGRU model with the physically-based model has proven particularly beneficial. This benefit arises from the WGRU model's ability to better handle the nonlinear relationships inherent in the simulated input data, which is especially surprising given that the standalone WGRU model initially showed the lowest performance. The hybrid mod-

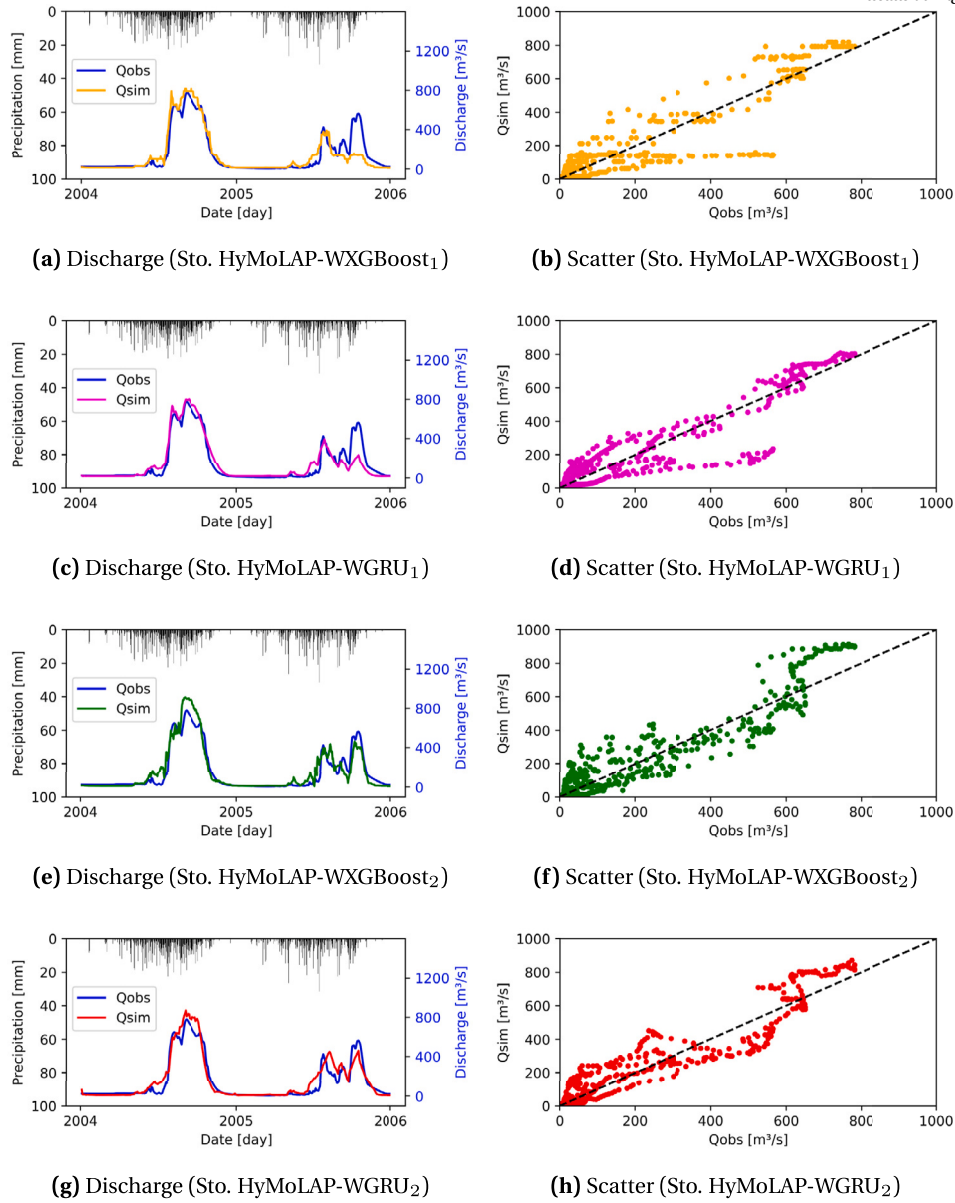


Fig. 5. Comparison of observed and simulated discharge, derived from the Sto. HyMoLAP-WXGBoost and Sto. HyMoLAP-WGRU hybrid models in the validation phase.

els in this study exclusively utilized simulated mean daily discharges along with information on daily quantiles. Considering the nature of this approach, we emphasize that integrating simulated statistical properties of daily discharge distributions from hydrological models into machine learning frameworks can significantly improve the accuracy of streamflow predictions. In other words, incorporating information about uncertainties related to rainfall-runoff systems can improve the point simulation of daily discharges, as demonstrated in this study. In the following subsections, we analyze errors across discharge classes, and examine the impact of the various uncertainty range in the second scenario on model performance.

3.3. Error analysis by discharge classes and performance evaluation

This subsection presents the error analysis of the HyMoLAP and different hybrid models, considering different classes of discharge such as discharge less than or equal to the 25th percentile, discharge between the 25th and 75th percentiles, and discharge greater than the 75th percentile. Therefore, we can define the discharge classes as follows: low

discharge (≤ 25 th perc.), medium discharge (25th - 75th perc.), and high discharge (> 75 th perc.). The analysis employs boxplots and the Root Mean Square Error (RMSE) to evaluate the predictive performance of each model across these discharge classes based on the error, as shown in Fig. 8. We chose RMSE for this analysis because it gives more weight to larger errors (by squaring them). As a result, it is more sensitive to outliers, unlike the MAE. The box plots provide a visual representation of the error distribution across the various discharge classes. The 25th and 75th percentiles correspond to $22.125 \text{ m}^3/\text{s}$ and $210.225 \text{ m}^3/\text{s}$, respectively.

The results indicate that all the hybrid models outperform the HyMoLAP model under low discharge conditions, demonstrating the hybrid model's strong capability to improve the simulation of low discharge events. In contrast, for medium discharges, the HyMoLAP-WXGBoost₁ and HyMoLAP-WGRU₁ models show the best performance. This shows that without incorporating quantile information, using only the average simulated discharge by the HyMoLAP model as input helps improve accuracy for medium discharges. This observation can also be attributed to the hybridization of the deterministic HyMoLAP

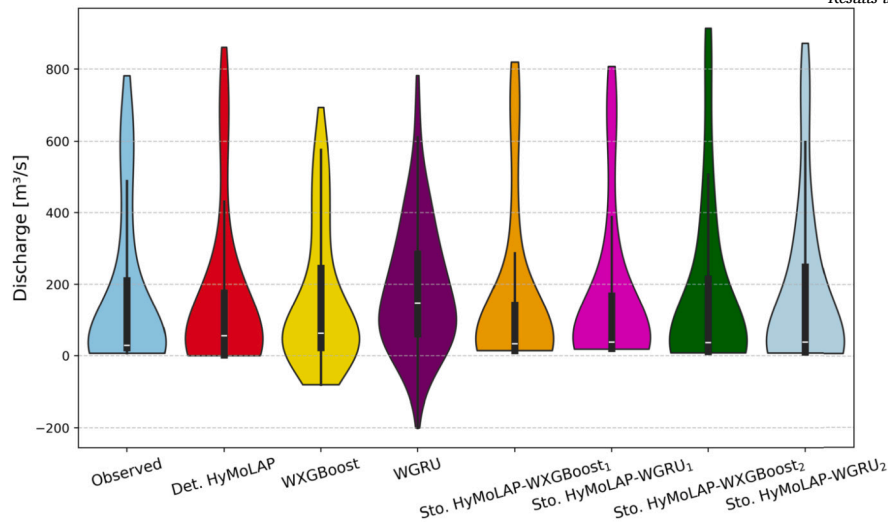


Fig. 6. Violin charts of predicted runoff using HyMoLAP, machine learning, and hybrid models.

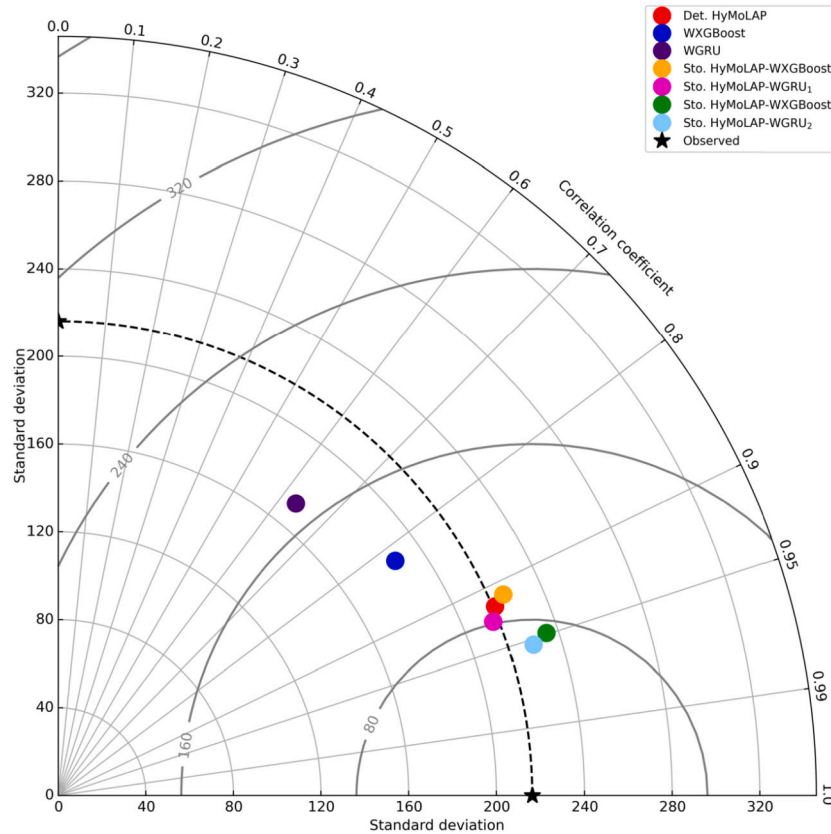


Fig. 7. Taylor diagram showing the performance of HyMoLAP, machine learning, and hybrid models for runoff prediction.

model with machine learning techniques, as the only input here is the average trajectory, and it has been observed above that, on average, the stochastic model produces performance similar to that of its deterministic form. Additionally, Sto. HyMoLAP-WGRU₂ demonstrates a slight improvement over HyMoLAP predictions. However, high discharges present a greater challenge for the models. As shown, HyMoLAP-WGRU₂ delivers the best performance in simulating high discharges, followed by HyMoLAP-WXGBoost₂. Unfortunately, HyMoLAP-WXGBoost₁, which performs well for low and medium discharges, performs the worst for high discharges—even when compared to the standalone HyMoLAP model. This limitation in capturing high discharges diminishes its overall predictive capability.

In summary, HyMoLAP-WGRU₂, which shows the best overall improvement (as presented in Table 6), also achieves significant enhancements over the standalone HyMoLAP when considering different discharge classes. Notably, it excels at simulating low and high discharges, which are crucial for studying extreme hydrological events. The ability to effectively improve the simulation of these extreme values is essential for flood forecasting, drought prediction, and the assessment of risks associated with extreme weather conditions. By enhancing the representation of such events, the model boosts the reliability of hydrological predictions under varying conditions, thereby supporting more informed decision-making in water resource management and disaster risk mitigation.

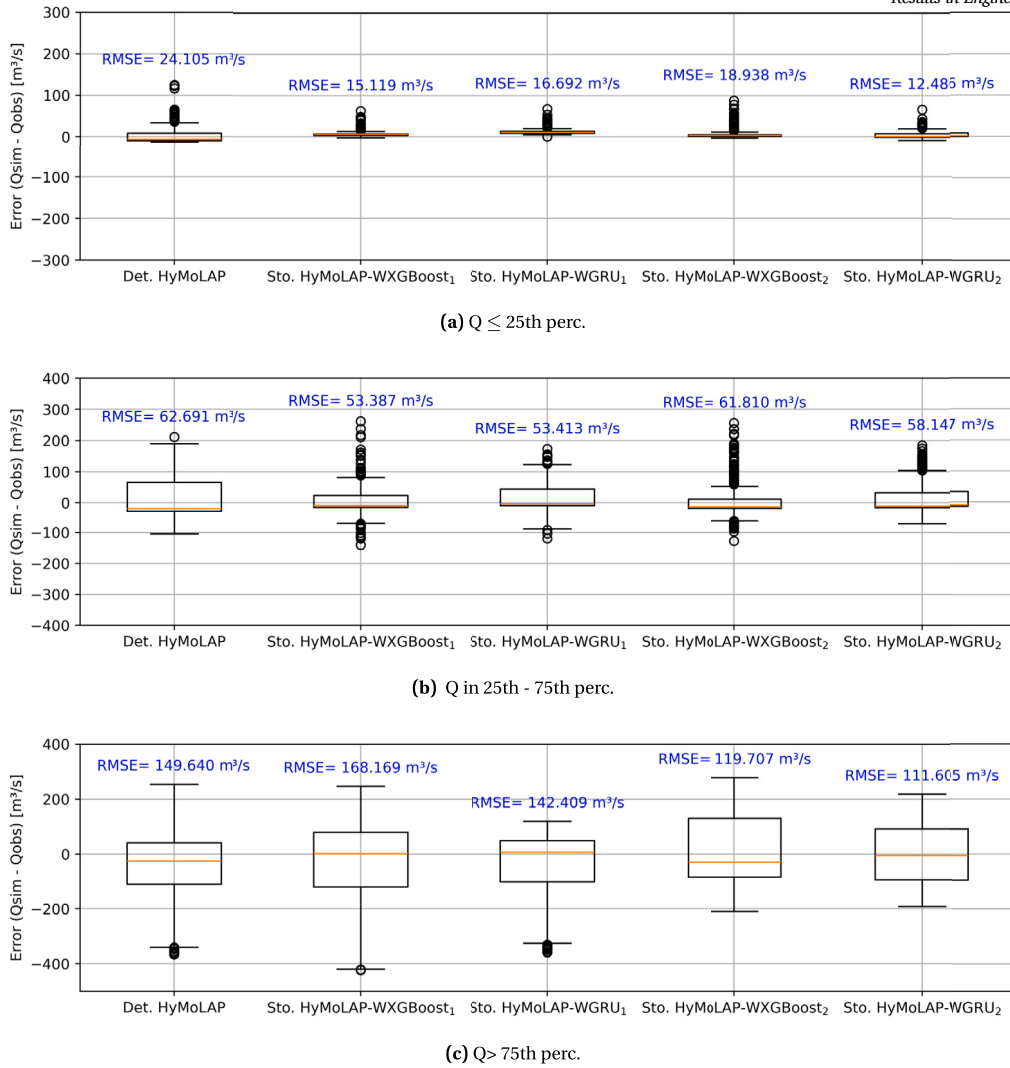


Fig. 8. Error analysis by discharge classes and comparison of observed and simulated discharge, of the standalone HyMoLAP and the hybrid models in the validation phase.

3.4. Sensitivity analysis of uncertainty ranges

The sensitivity analysis of uncertainty ranges in this study aims to understand how the consideration of different ranges of uncertainties associated with the rainfall-runoff system, as modeled by the Lévy-driven stochastic model (1), can contribute to enhancing point-based simulations of these discharges. To ensure robust statistical estimation of the daily probability density function's properties, the number of trajectories N_{traj} in the Sto. HyMoLAP simulation was set to 300,000. This high value helps to capture associated variability and uncertainties effectively, avoiding significant fluctuations with different sets of N_{traj} simulated trajectories.

Fig. 9 presents the evolution of the NSE criteria considering different uncertainty ranges or quantile levels (qL) for the Sto. HyMoLAP-WXGBoost₂ and Sto. HyMoLAP-WGRU₂ hybrid models. We limited the quantile levels to $q_L = 0.25$, corresponding to a 50% Quantile Interval (spanning from 25th to 75th percentile), which captures the central half of the data distribution around the median. Extending up to $q_L = 0.5$ would reduce the interval width, ultimately reaching only the median and thus excluding information about the distribution spread. Such a limitation would not align with the objectives of this analysis, as it would provide no insight into the uncertainties. This limitation is also related to the complexity of training both models for each quantile level, especially for the Sto. HyMoLAP-WGRU₂ model. Additionally, during

the simulations, a step size of 0.001 for qL was chosen for the Sto. HyMoLAP-WXGBoost₂, whereas a larger step size of 0.01 was used for the Sto. HyMoLAP-WGRU₂.

For the Sto. HyMoLAP-WXGBoost₂ hybrid model, the results show that at quantile level $q_L = 0$, corresponding to the full range of simulated uncertainties, the predictions are not very accurate, and less accurate than the standalone HyMoLAP model. However, prediction accuracy gradually increases and reaches a maximum level of 0.881 at $q_L = 0.027$. Beyond this threshold, performance begins to decline and tends to stabilize around 0.83 up to $q_L = 0.25$. The poor performance observed at the beginning ($q_L = 0$) could be linked to the influence of Lévy noise. At the quantile level $q_L = 0$, the full range of simulated uncertainties is taken into account, including the simulated large values which may deviate from actual observations. This can result in significant dispersion in the simulated trajectories by the Sto. HyMoLAP, which, when considering the full range, diminishes the accuracy of the hybrid models in predicting point discharge and consequently leads to a lower NSE. As the quantile level increases, the more extreme values in the simulated trajectories are excluded, which could explain the gradual improvement in performance, as the model focuses on values closer to actual observations. This reduction in extreme uncertainties leads to more accurate predictions up to a certain point ($q_L = 0.027$), before the opposite effect occurs, where the further reduction in information

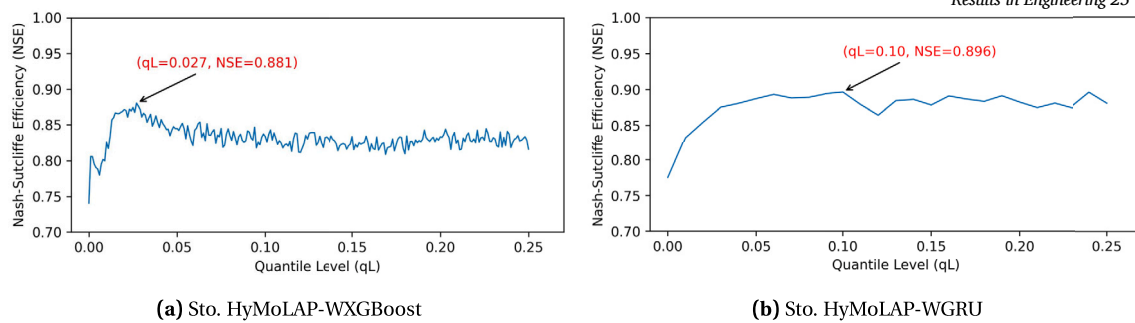


Fig. 9. Evolution of the Nash-Sutcliffe Efficiency (NSE) considering different uncertainty ranges for the Sto. HyMoLAP-WXGBoost₂ and Sto. HyMoLAP-WGRU₂ hybrid models in the validation phase.

can reduce performance by excluding important information about the variability of the system.

Considering the full range of uncertainties simulated by the Sto. HyMoLAP model, the same observation can also be made in the Sto. HyMoLAP-WGRU₂ hybrid model, where the accuracy is weak. However, the Sto. HyMoLAP-WGRU₂ hybrid model shows a certain stability in its performance, after reaching a plateau of 0.896 at $qL=0.10$. This indicates that it is less sensitive to variations in quantile levels compared to the Sto. HyMoLAP-WXGBoost₂ hybrid model. This stability can be explained by the fact that GRUs tend to be more robust to changes in quantile levels, consistently integrating these variations over time without significantly disrupting the model's overall performance. Additionally, the GRU's ability to learn temporal dependencies in the data may help smooth out short-term fluctuations, making the model more resilient to changes in quantile levels, thereby leading to a more stable and reliable prediction across a broader range of uncertainty scenarios.

As highlighted in [83], some river basins exhibit flow with highly complex behavior. This complexity often arises from random fluctuations in their rainfall-runoff processes. Simulating discharge, particularly over the long term, using only precipitation and potential evapotranspiration data can be particularly challenging in such cases. The findings of this study, from both hybrid model cases, suggest that the proposed hybridization technique could offer significant advantages for runoff simulations in catchments characterized by substantial random fluctuations. Furthermore, the analysis presented here shows that this hybrid approach can be applied even when using other stochastic methods for quantifying uncertainty in hydrological models, such as Gaussian methods or even Bayesian approaches.

4. Conclusion

A novel hybridization approach for rainfall-runoff modeling is presented in this study. The introduced approach predicts discharge solely based on observed precipitation and evapotranspiration data, without relying on previously observed discharge data. The hybridization integrates the physically-based stochastic HyMoLAP model with machine learning methods, such as Wavelet-based eXtreme Gradient Boosting (WXGBoost) and Wavelet-based Gated Recurrent Unit (WGRU). The Discrete Wavelet Transform (DWT) was applied to extract critical information from the input data, thereby enhancing the accuracy of the machine learning-based models. This hybrid approach integrates insights into simulated discharge-related uncertainties within the machine learning frameworks, thereby improving the accuracy of point discharge predictions. The study used only daily average discharge and various quantiles, estimated through Monte Carlo simulation of the physically-based stochastic model, from among the statistical properties of the discharge probability density function.

The results indicate that the standalone machine learning models provide the lowest accuracy across various evaluation metrics, including Nash–Sutcliffe efficiency (NSE), coefficient of determination (R^2), root mean square error (RMSE), Mean Absolute Error (MAE), and Combined

Accuracy (CA) when applied to the Ouémé at Bonou river basin, highlighting the limited ability of machine learning to capture patterns from only observed precipitation and potential evapotranspiration data to predict streamflow. On the other hand, hybrid models outperform both standalone physically-based and machine learning models. The sensitivity analysis of uncertainty ranges shows that the WGRU-based hybrid model exhibits better stability and superior performance compared to the WXGBoost-based hybrid model, particularly at higher quantile levels (i.e., narrower quantile intervals). This difference can be attributed to the GRU's ability to better handle the complex temporal dynamics in the simulated input data used in this study, making it more suitable for the hybridization approach. Such an approach could be particularly useful for runoff simulations in catchments where point discharge simulation is challenging. These findings have practical implications for enhancing the ability to prevent extreme hydrological events.

Overall, the results strongly support the hybridization approach, which can be applied with any model exhibiting stochastic behavior. However, its performance is highly dependent on the properties of the simulated discharge distribution used as input for the machine learning models. This underscores a key limitation: not all simulated uncertainty characteristics generated by hydrological models may be relevant as input. Therefore, testing multiple characteristics is essential to identify the most important ones for each specific case study. The study acknowledges the potential of advanced machine learning methods and suggests that future research could explore integrating techniques such as Convolutional Neural Networks (CNN), LSTM-ALO, LSTM-INFO, RVM-IMRFO, ANFIS-WCAMFO, ELM-IRSA, RVM-DMOA, or ensemble models to further enhance the predictive capabilities of this hybrid approach. We also encourage future research that extends this hybrid approach to improve uncertainty quantification.

CRediT authorship contribution statement

Sianou Ezékiel Houénafa: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Olatunji Johnson:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Erick K. Ronoh:** Writing – review & editing, Validation, Supervision, Investigation. **Stephen E. Moore:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code developed here is available in the corresponding GitHub repository: https://github.com/Ezesia-lab/Sto-HyMoLAP_Hybrid/tree/293e138bf1d22b86f3350eb5868e91a2b5205370/Hybrid. Any additional data supporting the findings of this study are available from the authors upon reasonable request.

References

- [1] F. Kratzert, D. Klotz, C. Brenner, K. Schulz, M. Herrnegger, Rainfall–runoff modelling using long short-term memory (lstm) networks, *Hydrol. Earth Syst. Sci.* 22 (2018) 6005–6022.
- [2] G.S. Nearing, C.S. Pelissier, F. Kratzert, D. Klotz, H.V. Gupta, J.M. Frame, A.K. Sampson, Physically informed machine learning for hydrological modeling under climate nonstationarity, 2019.
- [3] G.S. Nearing, F. Kratzert, A.K. Sampson, C.S. Pelissier, D. Klotz, J.M. Frame, C. Prieto, H.V. Gupta, What role does hydrological science play in the age of machine learning?, *Water Resour. Res.* 57 (2021) e2020WR028091.
- [4] C. Sezen, M. Šraj, Improving the simulations of the hydrological model in the karst catchment by integrating the conceptual model with machine learning models, *Sci. Total Environ.* 926 (2024) 171684.
- [5] A. Kapoor, S. Pathiraja, L. Marshall, R. Chandra, Deepgr4j: a deep learning hybridization approach for conceptual rainfall-runoff modelling, *Environ. Model. Softw.* 169 (2023) 105831.
- [6] V. Kumar, S. Unal, S.K. Bhagat, T. Tiyyasha, A data-driven approach to river discharge forecasting in the Himalayan region: insights from aglar and paligaad rivers, *Results Eng.* 22 (2024) 102044.
- [7] D.L. Shrestha, Uncertainty analysis in rainfall-runoff modelling-application of machine learning techniques, UNESCO-IHE PhD thesis, 2010.
- [8] U. Perera, D. Coralage, I. Ekanayake, J. Alawaturgoda, D. Meddage, A new frontier in streamflow modeling in ungauged basins with sparse data: a modified generative adversarial network with explainable ai, *Results Eng.* 21 (2024) 101920.
- [9] S. Asadi, P. Jimeno-Sáez, A. López-Ballesteros, J. Senent-Aparicio, Comparison and integration of physical and interpretable ai-driven models for rainfall-runoff simulation, *Results Eng.* 24 (2024) 103048.
- [10] X. Liang, D.P. Lettenmaier, E.F. Wood, S.J. Burges, A simple hydrologically based model of land surface water and energy fluxes for general circulation models, *J. Geophys. Res.* 99 (1994) 14415–14428.
- [11] M.B. Abbott, J.C. Bathurst, J.A. Cunge, P.E. O’Connell, J. Rasmussen, An introduction to the European hydrological system—système hydrologique européen, “she”, 1: history and philosophy of a physically-based, distributed modelling system, *J. Hydrol.* 87 (1986) 45–59.
- [12] S.L. Neitsch, J.G. Arnold, J.R. Kiniry, J.R. Williams, Soil and water assessment tool theoretical documentation version 2009, Technical Report, Texas Water Resources Institute, 2011.
- [13] A. Afouda, E. Lawin, T. Lebel, C. Peugeot, L. Seguis, Modèle de transformation pluie-débit basé sur le principe de moindre action, in: IAHS-AISH Publication, 2005, pp. 129–137.
- [14] A. Afouda, E. Alamou, Modèle hydrologique basé sur le principe de moindre action (modhypma), *Ann. Sci. Agron.* 13 (2010).
- [15] E.A. Alamou, Application du Principe de Moindre Action à la Modélisation Pluie-débit, Ph.D. thesis, Faculté des Sciences et Techniques (FAST), Université d’Abomey-Calavi (UAC), 2011.
- [16] E. Noether, Invariante variationsprobleme, in: *Gesammelte Abhandlungen-Collected Papers*, Springer, 1983, pp. 231–239.
- [17] A. Afouda, E.A. Lawin, T. Lebel, A stochastic streamflow model based on a minimum energy expenditure concept, in: *Contemporary Problems in Mathematical Physics*, World Scientific, 2004, pp. 153–169.
- [18] E.I. Biao, Assessing the impacts of climate change on river discharge dynamics in oueme river basin (Benin, West Africa), *Hydrology* 4 (2017) 47.
- [19] C. Gaba, E. Alamou, A. Afouda, B. Diekkriiger, Improvement and comparative assessment of a hydrological modelling approach on 20 catchments of various sizes under different climate conditions, *Hydrol. Sci. J.* 62 (2017) 1499–1516.
- [20] C. Perrin, C. Michel, V. Andréassian, Improvement of a parsimonious model for streamflow simulation, *J. Hydrol.* 279 (2003) 275–289.
- [21] J. Seibert, Hbv light version 2. User’s manual, Stockholm University, 2005.
- [22] S.E. Houénafa, E.K. Ronoh, O. Johnson, S.E. Moore, Lévy-induced stochastic differential equation models in rainfall-runoff systems for assessing extreme hydrological event risks, PREPRINT (Version 1) available at Research Square, <https://doi.org/10.21203/rs.3.rs-4618006/v1>, 2024.
- [23] S.M. Hauswirth, M.F. Bierkens, V. Beijl, N. Wanders, The potential of data-driven approaches for quantifying hydrological extremes, *Adv. Water Resour.* 155 (2021) 104017.
- [24] R.M. Adnan, A. Petroselli, S. Heddad, C.A.G. Santos, O. Kisi, Comparison of different methodologies for rainfall–runoff modeling: machine learning vs conceptual approach, *Nat. Hazards* 105 (2021) 2987–3011.
- [25] J. Adamowski, H.F. Chan, S.O. Prasher, V.N. Sharda, Comparison of multivariate adaptive regression splines with coupled wavelet transform artificial neural networks for runoff forecasting in Himalayan micro-watersheds with limited data, *J. Hydroinform.* 14 (2012) 731–744.
- [26] M. Rahimzad, A. Moghaddam Nia, H. Zolfonoon, J. Soltani, A. Danandeh Mehr, H.H. Kwon, Performance comparison of an lstm-based deep learning model versus conventional machine learning algorithms for streamflow forecasting, *Water Resour. Manag.* 35 (2021) 4167–4187.
- [27] H.Y. Dalkilic, D. Kumar, P. Samui, B. Dixon, S.N. Yesilyurt, O.M. Katipoğlu, Application of deep learning approaches to predict monthly stream flows, *Environ. Monit. Assess.* 195 (2023) 705.
- [28] L. Zhou, L. Kang, A comparative analysis of multiple machine learning methods for flood routing in the Yangtze River, *Water* 15 (2023) 1556.
- [29] N. Kedam, D.K. Tiwari, V. Kumar, K.M. Khedher, M.A. Salem, River stream flow prediction through advanced machine learning models for enhanced accuracy, *Results Eng.* 22 (2024) 102215.
- [30] O. Kisi, H.M. Azamathulla, F. Cevat, C. Kulls, M. Kuhdaragh, M. Fuladipannah, Enhancing river flow predictions: comparative analysis of machine learning approaches in modeling stage-discharge relationship, *Results Eng.* 22 (2024) 102017.
- [31] Y. Essam, Y.F. Huang, J.L. Ng, A.H. Birima, A.N. Ahmed, A. El-Shafie, Predicting streamflow in Peninsular Malaysia using support vector machine and deep learning algorithms, *Sci. Rep.* 12 (2022) 3883.
- [32] S. Talukdar, S. Pal, Shahfahad, M.W. Naikoo, A. Parvez, A. Rahman, Trend analysis and forecasting of streamflow using random forest in the Punarbhaba River basin, *Environ. Monit. Assess.* 195 (2023) 153.
- [33] M. Rahimzad, A. Moghaddam Nia, H. Zolfonoon, J. Soltani, A. Danandeh Mehr, H.H. Kwon, Performance comparison of an lstm-based deep learning model versus conventional machine learning algorithms for streamflow forecasting, *Water Resour. Manag.* 35 (2021) 4167–4187.
- [34] E.B. Wegayehu, F.B. Muluneh, Short-term daily univariate streamflow forecasting using deep learning models, *Adv. Meteorol.* 2022 (2022) 1860460.
- [35] X. Yuan, C. Chen, X. Lei, Y. Yuan, R. Muhammad Adnan, Monthly runoff forecasting based on lstm–alo model, *Stoch. Environ. Res. Risk Assess.* 32 (2018) 2199–2212.
- [36] R.M. Adnan, R.R. Mostafa, H.-L. Dai, S. Heddad, A. Masood, O. Kisi, Enhancing accuracy of extreme learning machine in predicting river flow using improved reptile search algorithm, *Stoch. Environ. Res. Risk Assess.* 37 (2023) 3063–3083.
- [37] R.M. Adnan, R.R. Mostafa, H.-L. Dai, E. Mansouri, O. Kisi, M. Zounemat-Kermani, Comparison of improved relevance vector machines for streamflow predictions, *J. Forecast.* 43 (2024) 159–181.
- [38] R.M. Adnan, R.R. Mostafa, H.-L. Dai, S. Heddad, A. Kuriqi, O. Kisi, Pan evaporation estimation by relevance vector machine tuned with new metaheuristic algorithms using limited climatic data, *Eng. Appl. Comput. Fluid Mech.* 17 (2023) 2192258.
- [39] R.M. Adnan, R.R. Mostafa, A.R.M.T. Islam, O. Kisi, A. Kuriqi, S. Heddad, Estimating reference evapotranspiration using hybrid adaptive fuzzy inferencing coupled with heuristic algorithms, *Comput. Electron. Agric.* 191 (2021) 106541.
- [40] R.M.A. Ikram, R.R. Mostafa, Z. Chen, K.S. Parmar, O. Kisi, M. Zounemat-Kermani, Water temperature prediction using improved deep learning methods through reptile search algorithm and weighted mean of vectors optimizer, *J. Mar. Sci. Eng.* 11 (2023) 259.
- [41] L. Ni, D. Wang, J. Wu, Y. Wang, Y. Tao, J. Zhang, J. Liu, Streamflow forecasting using extreme gradient boosting model coupled with Gaussian mixture model, *J. Hydrol.* 586 (2020) 124901.
- [42] M. Tayyab, J. Zhou, X. Dong, I. Ahmad, N. Sun, Rainfall-runoff modeling at jinsha river basin by integrated neural network with discrete wavelet transform, *Meteorol. Atmos. Phys.* 131 (2019) 115–125.
- [43] P.J. Lakshmi, R.A. Apaza, A. Alkhayyat, H.A. Marhoon, A.A. Alameri, Hybrid wavelet-gene expression programming and wavelet-support vector machine models for rainfall-runoff modeling, *Water Sci. Technol.* 86 (2022) 3205–3222.
- [44] D.K. Tiwari, V. Kumar, A. Goyal, K.M. Khedher, M.A. Salem, Comparative analysis of data driven rainfall-runoff models in the kolar river basin, *Results Eng.* 23 (2024) 102682.
- [45] G.A. Corzo Perez, Hybrid models for hydrological forecasting: Integration of data-driven and conceptual modelling techniques, 2009.
- [46] H. Lange, S. Sippel, Machine learning applications in hydrology, in: *Forest-Water Interactions*, 2020, pp. 233–257.
- [47] Y. Tian, Y.-P. Xu, Z. Yang, G. Wang, Q. Zhu, Integration of a parsimonious hydrological model with recurrent neural networks for improved streamflow forecasting, *Water* 10 (2018) 1655.
- [48] B.A. Yifru, K.J. Lim, J.H. Bae, W. Park, S. Lee, A hybrid deep learning approach for streamflow prediction utilizing watershed memory and process-based modeling, *Hydrol. Res.* 55 (2024) 498–518.
- [49] U. Okkan, Z.B. Ersoy, A.A. Kumanlioglu, O. Fistikoglu, Embedding machine learning techniques into a conceptual model to improve monthly runoff simulation: a nested hybrid rainfall-runoff modeling, *J. Hydrol.* 598 (2021) 126433.
- [50] T. Dang, L. Zhu, Euler-Maruyama schemes for stochastic differential equations driven by stable Lévy processes with i.i.d. stable components, *arXiv preprint, arXiv:2402.12502*, 2024.
- [51] M. Haugh, Simulating stochastic differential equations, in: *Monte Carlo Simulation: Ieor E 4703*, 2004.
- [52] R.H. McCuen, Z. Knight, A.G. Cutter, Evaluation of the Nash–sutcliffe efficiency index, *J. Hydrol. Eng.* 11 (2006) 597–602.

- [53] F. Harrou, A. Dairi, A. Dorbane, Y. Sun, Enhancing wind power prediction with self-attentive variational autoencoders: a comparative study, *Results Eng.* 23 (2024) 102504.
- [54] Y.-F. Sang, A review on the applications of wavelet transform in hydrology time series analysis, *Atmos. Res.* 122 (2013) 8–15.
- [55] V. Nourani, A.H. Baghanam, J. Adamowski, O. Kisi, Applications of hybrid wavelet-artificial intelligence models in hydrology: a review, *J. Hydrol.* 514 (2014) 358–377.
- [56] B. Cannas, A. Fanni, L. See, G. Sias, Data preprocessing for river flow forecasting using neural networks: wavelet transforms and data partitioning, *Phys. Chem. Earth Parts A/B/C* 31 (2006) 1164–1171.
- [57] I. Daubechies, The wavelet transform, time-frequency localization and signal analysis, *IEEE Trans. Inf. Theory* 36 (1990) 961–1005.
- [58] T. Chen, C. Guestrin, Xgboost: a scalable tree boosting system, in: *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [59] H. Ahmetoglu, R. Das, A comprehensive review on detection of cyber-attacks: data sets, methods, challenges, and future research directions, *Internet of Things* 20 (2022) 100615.
- [60] Z.A. Riyadi, J.O. Olutoki, M. Hermana, A.H.A. Latif, I.B.S. Yogi, S.J.A. Kadir, Machine learning prediction of permeability distribution in the x field Malay basin using elastic properties, *Results Eng.* 24 (2024) 103421.
- [61] A.J. Rios, M.E.A.B. Seghier, V. Plevris, J. Dai, Explainable ensemble learning framework for estimating corrosion rate in suspension bridge main cables, *Results Eng.* 23 (2024) 102723.
- [62] Kaggle, Your machine learning and data science community, <https://www.kaggle.com/>, 2021. (Accessed 2 December 2021).
- [63] R. Szczepanek, Daily streamflow forecasting in mountainous catchment using xgboost, lightgbm and catboost, *Hydrology* 9 (2022) 226.
- [64] M. Amjad, I. Ahmad, M. Ahmad, P. Wróblewski, P. Kamiński, U. Amjad, Prediction of pile bearing capacity using xgboost algorithm: modeling and performance evaluation, *Appl. Sci.* 12 (2022) 2126.
- [65] J. Lee, B. Xia, Analyzing the dynamics between crude oil spot prices and futures prices by maturity terms: deep learning approaches to futures-based forecasting, *Results Eng.* 24 (2024) 103086.
- [66] A. Saaidi, A. Bichri, S. Abderafi, Efficient machine learning model to predict dynamic viscosity in phosphoric acid production, *Results Eng.* 18 (2023) 101024.
- [67] J.M. Frame, F. Kratzert, D. Klotz, et al., Deep learning rainfall-runoff predictions of extreme events, *Hydrol. Earth Syst. Sci.* 26 (2022) 3377–3392.
- [68] M. Kheimi, M. Almadani, M. Zounemat-Kermani, Stochastic (s [arima]), shallow (narnet, nar-gmdh, os-elm), and deep learning (lstm, stacked-lstm, cnn-gru) models, application to river flow forecasting, *Acta Geophys.* (2023) 1–15.
- [69] L. Zhou, L. Kang, A comparative analysis of multiple machine learning methods for flood routing in the Yangtze River, *Water* 15 (2023) 1556.
- [70] X.-H. Le, D.-H. Nguyen, S. Jung, M. Yeon, G. Lee, Comparison of deep learning techniques for river streamflow forecasting, *IEEE Access* 9 (2021) 71805–71820.
- [71] K. Cho, B. Van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using rnn encoder-decoder for statistical machine translation, *arXiv preprint*, arXiv:1406.1078, 2014.
- [72] S. Hochreiter, Long Short-Term Memory, *Neural Computation* MIT-Press, 1997.
- [73] A. Dayal, S. Bonthu, P. Saripalle, R. Mohan, et al., Deep learning for multi-horizon water level forecasting in krs reservoir, India, *Results Eng.* 21 (2024) 101828.
- [74] P. Parisouj, C. Jun, S.M. Bateni, E. Heggy, S.S. Band, Daily runoff forecasting using novel optimized machine learning methods, *Results Eng.* 24 (2024) 103319.
- [75] O. Eray, C. Mert, O. Kisi, Comparison of multi-gene genetic programming and dynamic evolving neural-fuzzy inference system in modeling pan evaporation, *Hydrol. Res.* 49 (2018) 1221–1233.
- [76] R.M. Adnan, Z. Liang, S. Heddam, M. Zounemat-Kermani, O. Kisi, B. Li, Least square support vector machine and multivariate adaptive regression splines for streamflow prediction in mountainous basin using hydro-meteorological data as inputs, *J. Hydrol.* 586 (2020) 124371.
- [77] D.N. Moriasi, M.W. Gitau, N. Pai, P. Daggupati, Hydrologic and water quality models: performance measures and evaluation criteria, *Trans. ASABE* 58 (2015) 1763–1785.
- [78] C. Barber, J.R. Lamontagne, R.M. Vogel, Improved estimators of correlation and r2 for skewed hydrologic data, *Hydrol. Sci. J.* 65 (2020) 87–101.
- [79] T.O. Hodson, Root mean square error (rmse) or mean absolute error (mae): when to use them or not, *Geosci. Model Dev. Discuss.* (2022) 1–10.
- [80] C. Sezen, T. Partal, The utilization of a gr4j model and wavelet-based artificial neural network for rainfall-runoff modelling, *Water Supply* 19 (2019) 1295–1304.
- [81] J.L. Hintze, R.D. Nelson, Violin plots: a box plot-density trace synergism, *Am. Stat.* 52 (1998) 181–184.
- [82] K.E. Taylor, Summarizing multiple aspects of model performance in a single diagram, *J. Geophys. Res., Atmos.* 106 (2001) 7183–7192.
- [83] N.N. Kourgialas, G.P. Karatzas, N.P. Nikolaidis, An integrated framework for the hydrologic simulation of a complex geomorphological river basin, *J. Hydrol.* 381 (2010) 308–321.